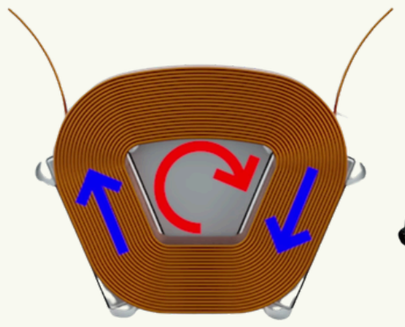
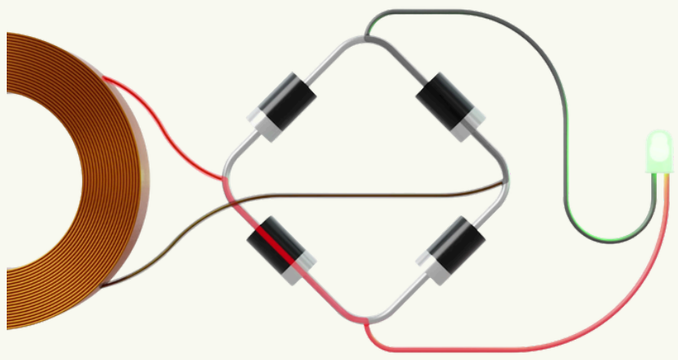
FLOOR TILE
ENERGY GENERATOR

NATHAN, ELBERT, RAEVEN,
DAEVIN, HARVEY



inspiration





HOW IT WORKS

- Pressure Activation:

When someone steps on the floor tile, a pressure plate is pushed down, generating mechanical energy.

- Cogwheel Motion:

This energy spins a cogwheel, which is connected to a set of rotating magnetic wheels.

- Magnetic Induction (Faraday's Law):

As the magnetic wheels spin, magnets move past coils of wire. According to Faraday's Law of Electromagnetic Induction, this movement of a magnetic field near a coil induces an electric current in the wire.

"The faster the magnets spin, the stronger the current and thus more electricity is generated."

Alternating Current (AC):
Because the magnets are arranged with alternating poles (North-South-North), the current produced switches direction—this is called alternating current (AC).

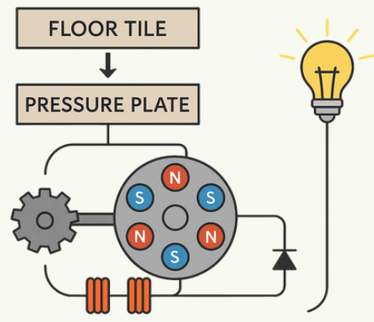
- Conversion to Direct Current (DC):

Most electronic devices like LED bulbs need direct current (DC). To convert AC to DC, a diode is placed in the circuit. The diode only allows current to flow in one direction, making the electricity safe and usable for the bulb.

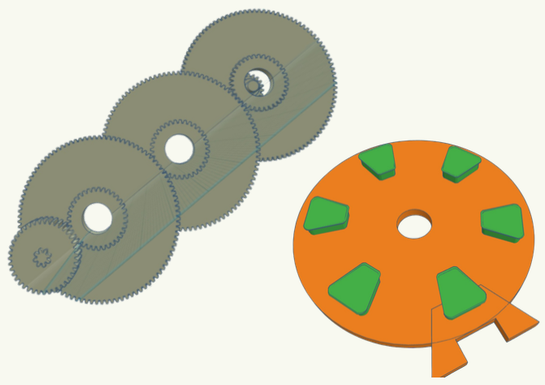
THEORY/CONCEPT

$$\mathcal{E} = -N\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

An energy-generating floor tile works by converting the pressure from a person's step into mechanical energy that spins magnets near coils of wire. This spinning creates electricity through magnetic induction (Faraday's Law). The current produced is alternating (AC) because of the magnet arrangement, but a diode is used to convert it into direct current (DC) so it can safely power devices like LED lights.



PROTOTYPE



EFFICIENCY & POWER OUTPUT

Assumptions:
Mass = 60Kg
Average Vertical velocity of the foot before it contacts the ground is about 0.2-0.4 m/s.
Velocity of the foot = 0.3m/s
Average time for the floor tile to fully sink down = 0.7s

Energy input:
E= 1/2mv^2
E = 1/2(60)(0.3)^2
E = 2.7J

Energy output:
Current (I) = 80 mA (approx)
Voltage (V) = 0.3V (approx)

P = IV
P = 80mA x 0.3V = 0.024W

P = E/t
0.024W = E/0.7
E = 0.0168 J
= 0.02 J

Efficiency Calculation:
η = (Energy Output / Energy Input) × 100%
η = 0.02 J /2.7J x 100%
η = 0.74%

ERROR ANALYSIS & FUTURE WORK

Mechanicals:
Mechanical stuff such as gears and the pully system produces friction which reduces the amount of motion that reaches the Magnet wheel. Gaps and instability in the mechanism might also cause energy loss.

Electromagnetics:
The magnet strength might not be strong enough because we already bought it from a long time ago and temperature change, impacts and pressure could affect the strength of the magnetic field.
The coils might also not have enough coil winding for it to create a strong current.

Current Conversion:
Currents conversion from the diodes may increase the resistance in the circuit due to diodes having high resistance.

Future Work:
How this project could be improved in the future is maybe by using a more reliable gear stabilizer and lower friction materials. And instead of using a pully system, maybe use spring or other mechanical retraction method that does not create high energy loss.
More coil windings and stronger magnets may also be applied for a better electromagnet energy conversion.

PROS & CONS

PROS :

- Renewable energy source – uses human movement, no fuel needed
- Eco-friendly – reduces carbon footprint
- Space-saving – works in places with high foot traffic (malls, airports, schools)

CONS :

- High cost – expensive to install and maintain
- Durability issues – moving parts may wear out over time
- Efficiency depends on usage – not effective in low-traffic areas